

Basic Questions to know:

1). what is Java Script?

JS is high-level, lightweight, interpreted programming language used to make web pages interactive and dynamic single thread process.

2). why was JS created?

Before JS created the web pages that were previously static HTML pages to add interactivity to the web pages using Javascript

ex: used for

→ respond for user actions

→ validation forms

→ update content without reloading pages.

3). Is JS Interpreted or compiled?

The Modern JS is both Interpreted and Just-in-Time (JIT) compiled.

but Earlier: Mostly it interpreted

Today: JS engine compile code into machine code before execution for better performance.

(JIT)

So, JS is compiled language that combined both interpreted and compilation for faster execution.

4). Features of JavaScript.

- It is lightweight
- Dynamic Typing
- Object-Oriented
- Event-Driven
- cross-Platform
- Single-Threaded
- asynchronous programming
- first-class function.

5). why JS called single-threaded language.

It executes one task at a time using a single call stack. (execute one call stack and one operation at a time).

6). diff between

JavaScript and Java.

JS

Java

Programming Language

Scripting Language

Dynamically Typed

Statically Typed

Runs in Browser &

Runs on JVM

~~Server~~

~~Prototype Based~~

~~Class Based~~

~~Lightweight~~

~~Heavier~~

used for web
Development

used for Enterprise
applications.

7). ECMA Script?

ECMAScript (ES) is official standard specification for JavaScript.

It defines:

- Syntax
- Rules
- Features

versions:

- ES5
- ES6
- ES7
- ES8

popular ES6 features:

- let
- const
- arrow function
- classes
- Promises

8). Role of JS Engine?

JS Engine:

- Reads JS code
- Parses code
- Converts it into machine code
- Executes machine code.

Responsibilities:

- Parsing
- Compilation
- Memory management
- Garbage collection
- Execution

9). V8 Engine?

JS engine that converts JS into machine code for fast execution.

- Fast execution
- JIT compilation
- Efficient memory management.

10). JS Executes Code:

